STA6166 Statistical Methods I

Final Project: Multiple Linear Regression and Variable Selection

Total: 100 Points

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Project Overview

You will analyze a dataset containing one continuous response variable Y and 15 candidate predictors X_1, …, X_{15}. The data were generated from a multiple linear regression model in which exactly 5 predictors have nonzero population regression coefficients; the other 10 have coefficients equal to zero. The identities of the 5 true predictors will not be provided.

Your tasks are to fit the full regression model, examine its assumptions, select predictors using one classical model-selection approach, fit a LASSO regression with a cross-validated tuning parameter, and compare the resulting models. Complete the analysis in R. Unless otherwise stated, use a 5% significance level.

Part 1. Full Multiple Linear Regression and Diagnostics (20 Points)

Question 1. Fit the full model (5 points). Fit the model $Y_i = \beta_0 + \sum_{j=1}^{15} \beta_j X_j + \epsilon_i$ Report the coefficient estimates, standard errors, t-statistics, and p-values. Include the model's R^2 and adjusted R^2.

Fill in the following table using the R output from the full model with 15 predictors.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	**1.9115**	**0.7175**	**2.664**	**0.009253**
β_2	0.2025	0.6982	0.290	0.772483
β_3	-0.3949	0.6771	-0.583	0.561335
β_4	**-2.6473**	**0.8055**	**-3.287**	**0.001481**
β_5	1.1445	0.7498	1.526	0.130681
β_6	0.9065	0.7347	1.234	0.220698
β_7	1.2767	0.6619	1.929	0.057121
β_8	0.2487	0.7648	0.325	0.745818
β_9	-0.1041	0.6638	-0.157	0.875731
β_{10}	**7.2157**	**0.7176**	**10.056**	**4.49e-16**
β_{11}	-0.1946	0.6258	-0.311	0.756631
β_{12}	0.3334	0.6508	0.512	0.609722
β_{13}	**-2.6675**	**0.6812**	**-3.916**	**0.000183**
β_{14}	**-1.6059**	**0.6871**	**-2.337**	**0.021801**
β_{15}	0.5023	0.6974	0.720	0.473400

Rows in bold are significant at α = 0.05.

$R^2 = 0.6506$ and $R^2_{\text{adj}} = 0.5882$. The overall F-test is highly significant ($F = 10.43$ on 15 and 84 DF, $p = 1.442 \times 10^{-13}$). At the 5% level, X1, X4, X10, X13 and X14 are significant predictors; X7 is borderline ($p = 0.057121$).

```
> summary(full_model)     # last lines of the output
Residual standard error: 6.427 on 84 degrees of freedom
Multiple R-squared:  0.6506,    Adjusted R-squared:  0.5882
F-statistic: 10.43 on 15 and 84 DF,  p-value: 1.442e-13
```

Question 2. Unequal error variances (5 points). Apply the Breusch–Pagan test to the full regression model.

(a) State the null and alternative hypothesis.

H_0: $\text{Var}(\varepsilon_i) = \sigma^2$ for all i (constant error variance, homoscedasticity).

H_1: $\text{Var}(\varepsilon_i)$ is not constant; it depends on the predictors $X_1, \ldots, X_{15}$ (heteroscedasticity).

(b) Write down the test statistic and p-value you get from the R function.

Test statistic $\text{BP} = 17.673$ with $\text{df} = 15$; p-value $= 0.2802$.

```
> print(bp_result)
    studentized Breusch-Pagan test

data:  full_model
BP = 17.673, df = 15, p-value = 0.2802
```

(c) State if you have enough evidence to reject H_0.

No. Since $p = 0.2802 > \alpha = 0.05$, we fail to reject H_0. There is not enough evidence of unequal error variances.

Question 3. Normality diagnostic (5 points). Apply the Shapiro–Wilk test to test the normality of residuals of the full regression model.

(a) State the null and alternative hypothesis.

H_0: the residuals come from a normal distribution.

H_1: the residuals do not come from a normal distribution.

(b) Write down the test statistic and p-value you get from the R function.

Test statistic $W = 0.99144$; p-value $= 0.7795$.

```
> print(shapiro_result)
    Shapiro-Wilk normality test

data:  res
W = 0.99144, p-value = 0.7795
```

(c) State if you have enough evidence to reject H_0.

No. Since $p = 0.7795 > \alpha = 0.05$, we fail to reject H_0. There is not enough evidence that the residuals depart from normality.

Question 4. Diagnostic plots (5 points). Provide (a) a residuals-versus-fitted plot and (b) a normal Q–Q plot. Briefly discuss whether the plots are consistent with constant error variance and normality.

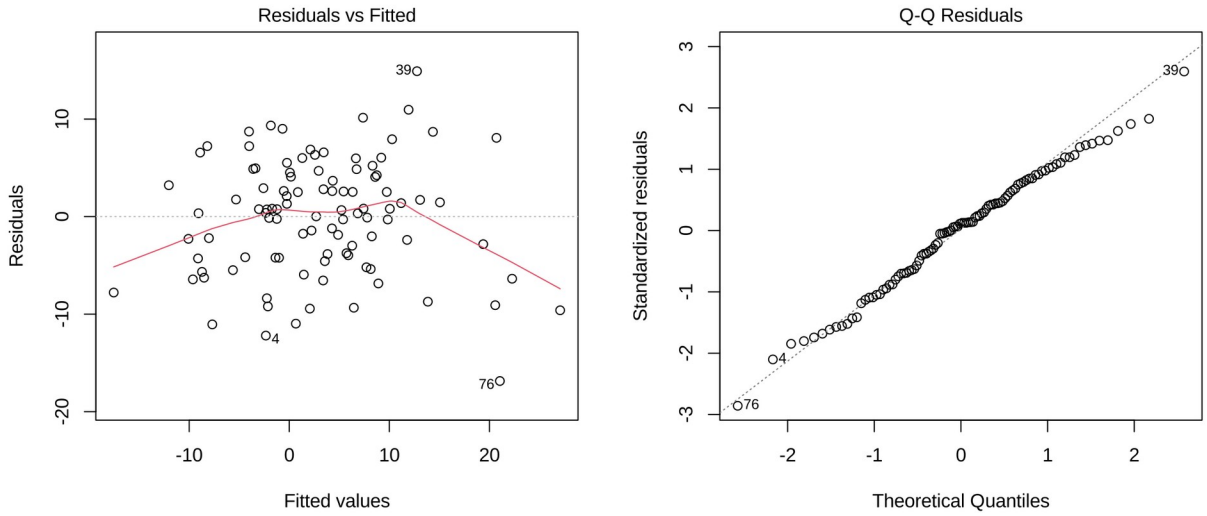


Figure 1. (a) Residuals versus fitted values and (b) normal Q–Q plot for the full model.

> **(a) Residuals vs fitted:** the residuals are scattered randomly around zero with roughly the same spread across all fitted values, and there is no funnel shape. The slight bend in the red line at the ends comes from a few extreme points (e.g., observation 76), not a systematic pattern. This is consistent with **constant error variance**.
>
> **(b) Normal Q–Q:** the points lie close to the reference line, with only small deviations in the tails (observations 39, 76 and 4). This is consistent with **normality**.
>
> Both plots agree with the Breusch–Pagan and Shapiro–Wilk results.

Part 2. Classical Variable Selection (30 Points)

Question 5. Choose a search procedure (5 points). Choose ONE procedure: forward selection, backward elimination, or bidirectional stepwise selection. Explain how the procedure (the ONE procedure you choose) adds or removes variables.

> **I chose bidirectional stepwise selection.**
>
> - It starts from the intercept-only model, $Y = \beta_0 + \varepsilon$.
> - At each step it considers adding each predictor not yet in the model and removing each predictor already in the model, and it makes the one move that improves the criterion most.
> - Because removal is allowed at every step, a predictor added earlier can be dropped later if it is no longer useful.
> - It stops when no single addition or removal improves the criterion.

Question 6. Choose one selection criterion (5 points). Choose exactly one criterion: AIC, BIC, Mallows' C_p, or adjusted R^2. Explain what the criterion measures and why you chose it. Once selected, keep this criterion fixed throughout Part 2; do not switch criteria after seeing the results.

For AIC or BIC, you may use stats::step() with forward, backward, or bidirectional search. For C_p or adjusted R^2, you may use leaps::regsubsets() with forward or backward search. The seqrep method in regsubsets() is sequential replacement, not bidirectional stepwise regression. If you choose an unsupported combination, you must implement and document the search algorithm yourself. You may revise your initial choice of search procedure before running the analysis to use one of the supported combinations.

> **I chose BIC (Bayesian Information Criterion).**
>
> $$\text{BIC} = -2\log L + p\log N = N\log\left(\frac{\text{RSS}}{N}\right) + p\log N + \text{constant}$$
>
> where L is the maximized likelihood, p is the number of estimated parameters and $N = 100$ is the sample size. The first term measures lack of fit and the second penalizes model complexity; the model with the **smallest BIC** is preferred.
>
> I chose BIC because its penalty per parameter, $\log(100) \approx 4.61$, is heavier than the AIC penalty of 2, so it favours simpler models. This suits the project, where only 5 of the 15 predictors are truly nonzero. BIC with bidirectional search is supported by stats::step(), using $k = \log N$.

```
> stepwise_BIC <- step(null_model,
+   scope = list(lower = formula(null_model), upper = formula(full_model)),
+   direction = "both",
+   k = log(N), trace = 0)
```

Question 7. Select predictors (10 points). Apply the chosen search procedure and fixed criterion.

(a) List the predictors in the final selected model.

> **X1, X4, X10, X13 and X14.**

(b) Report the final model formula.

> **Y ~ X10 + X4 + X13 + X14 + X1**, that is,
>
> $$\hat{Y} = 1.9307 + 1.6604\,X_1 - 2.6507\,X_4 + 7.2586\,X_{10} - 2.3389\,X_{13} - 1.8656\,X_{14}$$

```
> formula(stepwise_BIC)
Y ~ X10 + X4 + X13 + X14 + X1
```

(c) Report the chosen criterion's value for the selected model.

> BIC = 680.8865

```
> BIC(stepwise_BIC)
[1] 680.8865
```

(d) State the number of selected predictors, excluding the intercept. Briefly describe how you identified the final model along the search path.

> **5 predictors** (excluding the intercept).
>
> Starting from the intercept-only model, step() added the predictors one at a time in the order they appear in the formula: X10, X4, X13, X14 and then X1. At each step it also checked whether removing a predictor would lower BIC, but none was removed. The search stopped when no further addition or removal lowered BIC, and that model was taken as the final model. The forward and backward BIC searches (sections 6.4 and 6.5 of the code) gave the same five predictors.

Question 8. Examine the selected model (10 points). Refit an ordinary least-squares model using the selected predictors.

(a) Fill in the following table using the R output from the model with selected predictors. If a variable is not selected, leave it empty.

Coefficient	Point estimate	Standard error	t-statistic	p-value
β_1	1.6604	0.6892	2.409	0.017936
β_2				
β_3				
β_4	-2.6507	0.7873	-3.367	0.001103
β_5				
β_6				
β_7				
β_8				
β_9				
β_{10}	7.2586	0.6732	10.782	< 2e-16
β_{11}				
β_{12}				
β_{13}	-2.3389	0.6343	-3.687	0.000379
β_{14}	-1.8656	0.6545	-2.850	0.005366
β_{15}				

(b) Report adjusted R^2.

> $R^2_{adj} = 0.5925$

```
> summary(stepwise_BIC)     # last lines of the output
Residual standard error: 6.393 on 94 degrees of freedom
Multiple R-squared:  0.6131,    Adjusted R-squared:  0.5925
F-statistic: 29.79 on 5 and 94 DF,  p-value: < 2.2e-16
```

(c) Compare its adjusted R^2 with that of the full model.

> The selected model has $R^2_{\text{adj}} = 0.5925$, slightly **higher** than the full model's $R^2_{\text{adj}} = 0.5882$, even though it uses 10 fewer predictors. The extra predictors in the full model did not improve the fit enough to justify their cost.

(d) Discuss whether the selected model is simpler and what information may have been lost by excluding predictors.

> **Yes, the selected model is simpler:** 5 predictors instead of 15, and all five are significant at the 5% level (largest $p = 0.017936$).
>
> **Information lost:** R^2 falls from 0.6506 to 0.6131, so the excluded predictors explained about 3.75% of the variation in Y. If any of them has a small real effect, it is no longer in the model. X7 ($p = 0.057$ in the full model) and X5 are the most likely examples, since AIC and Mallows' C_p kept both.

Part 3. LASSO Regression (30 Points)

Question 9. Explain LASSO (5 points). Consider the LASSO objective: min over β_0, β of $\frac{1}{2N}\sum_{i=1}^{N}\left(Y_i-\beta_0-\sum_{j=1}^{15}X_{ij}\beta_j\right)^2+\lambda\sum_{j=1}^{15}|\beta_j|$.

(a) Explain the role of λ.

$\lambda \geq 0$ is the **tuning parameter** that controls the strength of the penalty $\lambda\sum_{j=1}^{15}|\beta_j|$.

- When $\lambda = 0$ there is no penalty and LASSO gives the ordinary least-squares estimates.
- As λ increases, the coefficients are shrunk more toward zero and more of them become exactly zero.
- When λ is very large, all slope coefficients $\beta_1,\ldots,\beta_{15}$ are zero.

So λ balances fitting the data well against keeping the model simple.

(b) Explain why LASSO can produce exactly zero coefficient estimates. The intercept is not penalized.

The penalty uses absolute values $|\beta_j|$, which have a sharp corner at zero. Because of this corner, the best solution for a weak predictor is often exactly $\hat{\beta}_j = 0$. With standardized predictors, each LASSO estimate is a soft-thresholded value:

$$\hat{\beta}_j = \text{sign}(z_j)\max(|z_j| - \lambda, 0)$$

where z_j measures the association between X_j and the current residuals. Whenever $|z_j| \leq \lambda$, the estimate is set exactly to zero rather than just made small. Geometrically, the constraint region $\sum_{j=1}^{15}|\beta_j| \leq t$ is a diamond with corners on the axes, and the solution often falls on a corner, where some coefficients are zero. The intercept β_0 is not penalized, so it is never shrunk.

Question 10. Choose λ by cross-validation (10 points). Use glmnet::cv.glmnet() with 10-fold cross-validation and Gaussian squared-error loss. Use a fixed random seed for reproducibility and standardize predictors using the package's default behavior.

(a) Include the cross-validation plot, plot the cross-validation MSE over λ.

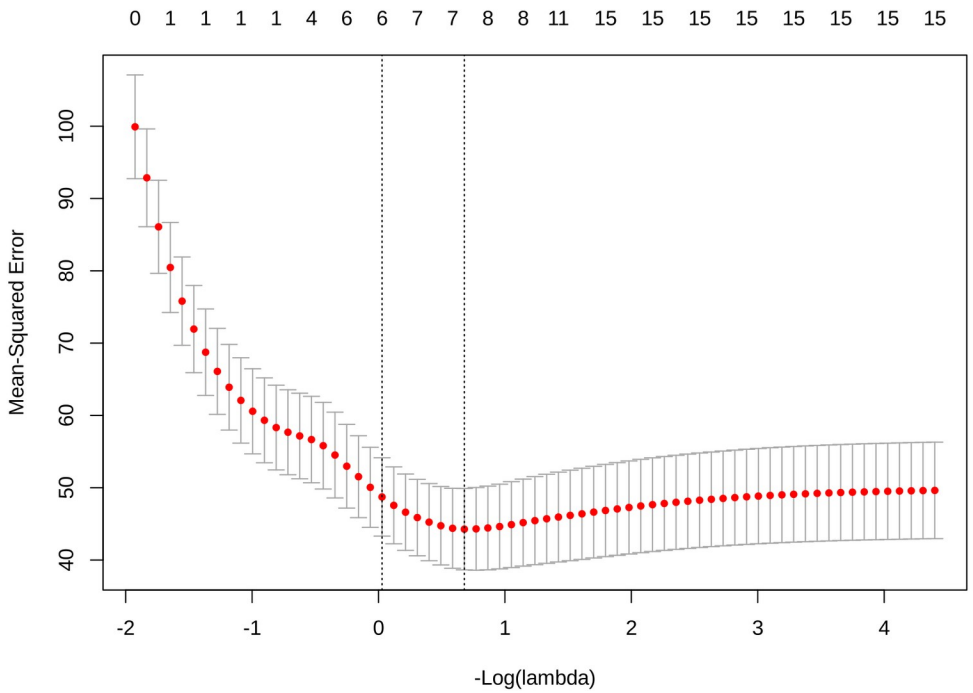


Figure 2. 10-fold cross-validation MSE versus −log(λ) (λ decreases from left to right). The right dashed line marks lambda.min and the left one lambda.1se; the numbers on top are the number of nonzero coefficients.

(b) Report lambda.min, the λ giving the smallest cross-validation MSE.

> $\lambda_{min} = 0.5070394$ (lambda.min), giving the smallest cross-validation MSE of 44.2725.

```
> print(cv_lasso)

Call:  cv.glmnet(x = X, y = Y, type.measure = "mse", nfolds = 10, alpha = 1,
       family = "gaussian", standardize = TRUE)

Measure: Mean-Squared Error

    Lambda Index Measure    SE Nonzero
min 0.5070    29   44.27 5.628       8
1se 0.9725    22   48.73 5.419       6

> lambda_selected
[1] 0.5070394
```

(c) Explain how the cross-validation criterion selects λ. Use lambda.min throughout the remaining LASSO analysis; do not switch to lambda.1se.

> The data are randomly split into 10 folds (seed 6166). For each value of λ, the LASSO model is fitted on 9 folds and its mean squared error $\mathrm{MSE}_k(\lambda)$ is computed on the left-out fold k. Repeating this so that each fold is left out once and averaging gives the cross-validation error:
>
> $$\mathrm{CV}(\lambda)=\frac{1}{10}\sum_{k=1}^{10}\mathrm{MSE}_k(\lambda),\lambda_{\min}=\arg\min_{\lambda}\mathrm{CV}(\lambda)$$
>
> The λ with the **smallest average MSE** is lambda.min. A λ that is too small overfits the data, and one that is too large shrinks the coefficients too much, so the minimum lies in between. lambda.min is used for the rest of the analysis.

Question 11. Fit LASSO and identify variables (10 points). Using the selected λ, obtain the LASSO coefficient estimates.

(a) Report the coefficients.

Enter the LASSO point estimate for each predictor. If LASSO does not select a predictor, enter 0.

Coefficient	LASSO point estimate
β_1	**1.07944426**
β_2	0.00000000
β_3	0.00000000
β_4	**-1.92849719**
β_5	**0.41838736**
β_6	**0.06818692**
β_7	**0.89926976**
β_8	0.00000000
β_9	0.00000000
β_{10}	**6.62840784**
β_{11}	0.00000000
β_{12}	0.00000000
β_{13}	**-1.62217204**
β_{14}	**-1.16931949**
β_{15}	0.00000000

(b) List all predictors with nonzero coefficients.

> **X1, X4, X5, X6, X7, X10, X13, X14**

```
> print(selected_predictors)
[1] "X1"  "X4"  "X5"  "X6"  "X7"  "X10" "X13" "X14"
```

(c) State how many predictors LASSO selected. Exclude the intercept from the count.

> LASSO selected **8 predictors**.

```
> length(selected_predictors)
[1] 8
```

Question 12. Interpret LASSO (5 points). Explain how the penalty affects the fitted coefficients and why some predictors are excluded. Discuss the relationship between the size of λ and model sparsity.

> **Effect on the coefficients:** the penalty shrinks the coefficients toward zero compared with ordinary least squares. For example, $\hat{\beta}_{10}$ drops from 7.2157 (full model) to 6.6284, and $\hat{\beta}_{13}$ from -2.6675 to -1.6222. Weak predictors shrink the most: $\hat{\beta}_{6}$ drops from 0.9065 to only 0.0682.
>
> **Why some are excluded:** a predictor whose contribution to reducing the squared error is smaller than the penalty is set exactly to zero ($\hat{\beta}_j = 0$). Here X2, X3, X8, X9, X11, X12 and X15 were excluded; they were all far from significant in the full model.
>
> **λ and sparsity:** the larger λ is, the sparser the model. The numbers on top of Figure 2 show the number of nonzero coefficients falling from 15 at small λ, to 8 at lambda.min, 6 at lambda.1se, and 0 at the largest λ.

Part 4. Comparison and Discussion (20 Points)

Question 13. Compare selected predictors (10 points). Create a table with one row for each of X_1, …, X_{15} and two columns indicating whether each predictor was selected by (i) classical variable selection and (ii) LASSO. Report (a) the number selected by each method, (b) the predictors selected by both, and (c) those selected by only one method.

Predictor	Classical selection (Yes/No)	LASSO (Yes/No)
X1	**Yes**	**Yes**
X2	No	No
X3	No	No
X4	**Yes**	**Yes**
X5	No	**Yes**
X6	No	**Yes**
X7	No	**Yes**
X8	No	No
X9	No	No
X10	**Yes**	**Yes**
X11	No	No
X12	No	No
X13	**Yes**	**Yes**
X14	**Yes**	**Yes**
X15	No	No

(a) Classical selection (stepwise BIC) selected **5** predictors; LASSO selected **8** predictors.

(b) Selected by both: **X1, X4, X10, X13, X14**.

(c) Selected by only one method: **X5, X6, X7** (LASSO only). No predictor was selected only by the classical method.

Question 14. Explain differences (5 points). Do the methods select the same predictors? Which selects fewer? Explain at least two reasons the selected sets may differ, considering the criteria, coefficient shrinkage, and correlation among predictors.

No. Both methods select X1, X4, X10, X13 and X14, but LASSO also selects X5, X6 and X7. The **classical method (BIC) selects fewer** predictors (5 vs 8). Reasons the sets differ:

- **Different criteria:** BIC has a heavy penalty for each extra predictor and aims for a simple model, whereas lambda.min minimizes cross-validation prediction error, which often keeps a few weak predictors. In the code, AIC and C_p also kept 7 predictors and adjusted R^2 kept the same 8 as LASSO.
- **Coefficient shrinkage:** LASSO can keep a weak predictor with a very small coefficient (e.g., $\hat{\beta}_6 = 0.068$), while classical selection must include a predictor at its full OLS estimate or drop it completely.
- **Correlation among predictors:** when predictors are correlated, they share information. Stepwise selection may keep one and drop the other, while LASSO may keep both with smaller coefficients, which leads to different selected sets.

Question 15. Final discussion (5 points). State which method you prefer for this dataset and justify your choice using model simplicity, the selected predictors, and the results of your analysis. Do not assume that a selected predictor is necessarily a true predictor: the identities of the 5 nonzero population coefficients are unknown to you.

> **I prefer the classical bidirectional stepwise model with BIC (X1, X4, X10, X13, X14).**
>
> - **Model simplicity:** it uses 5 predictors instead of 8, and all five are significant at the 5% level.
> - **Selected predictors:** its five predictors are exactly the ones both methods agree on. The three extra LASSO predictors have small coefficients ($\hat{\beta}_5 = 0.418$, $\hat{\beta}_6 = 0.068$, $\hat{\beta}_7 = 0.899$) and were not significant in the full model.
> - **Results of the analysis:** its $R^2_{\text{adj}} = 0.5925$ is higher than the full model's 0.5882, the forward, backward and stepwise BIC searches all gave the same model, and print(cv_lasso) shows that the more conservative lambda.1se would keep only 6 predictors.
>
> However, a selected predictor is not necessarily a true predictor, since the 5 true predictors are unknown. X7 is borderline and could be a weak real effect.

Submission Requirements

Submit only the completed Microsoft Word document (.docx) containing your answers to all project questions. No additional files are required.

Grading Summary

Component	Points
Part 1. Full regression and diagnostics	20
Part 2. Classical variable selection	30
Part 3. LASSO regression	30
Part 4. Comparison and discussion	20
Total	100